

COMPSYS 306 Artificial Intelligence and Machine Learning

Project Pt. 1 Report

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I. INTRODUCTION

The purpose of this project is to design and implement a machine learning application for traffic sign recognition. This involves the making of two types of machine learning models, a Multi-Layer Perceptron (MLP) and a Support Vector Machine (SVM). Both models are written in Python and will use the publicly available traffic signs dataset from Kaggle to train, validate and evaluate the models.

II. DATA AND APPLICATION DESCRIPTION

A. Dataset

The public traffic sign dataset given to us to use for this project contains 73,139 images which are split across 43 distinct categories, with a csv file that contains the metadata of the traffic sign labels. There are 450 images for the categories that contain the lowest number of images whereas the category with the largest contain 4920 images. All images are 32 by 32 resolution in full rgb colour. Looking into the images themselves, in all the categories, the images vary in lighting and contrast. This is from the images being taken during various times of the day and different weather conditions. Therefore, some images will be overblown with lighting whereas some other images will be extremely dark, making the sign barely perceptible. Having all this data is good for the model as it ensures that the model will be able to classify the traffic sign under a variety of conditions.

B. Application Description

The goal of this project is to create a MLP and SVM model for classifying traffic signs and compare the performance between them. This task has real-world applications, particularly in autonomous driving systems, where accurate traffic sign recognition will be needed. Autonomous vehicles systems will need this capability to detect speed limits, stop signs, and other road signals. This will enable them to make safe, informed decisions and respond accordingly.

III. DATA PRE-PROCESSING

A. Standardization/ Normalization

For both MLP and SVM models, had the resolution of the images scaled down to 28 by 28 and converted to grayscale. By reducing the resolution of the image there will be less data to process during training therefore reducing the amount time

needed to train the model. Grayscale the images further reduces the computational complexity as it reduces the dimensions of the image from 3 channels to 1, thus further reducing the time it takes to train. Both models were also flattened as SVM require a 2d array input and MLP dense layers expect a 1d input.

B. Train-Test Data Split

A test-train data split of 80% training data and 20% testing data. This ensures that the models will have many images to train from. As the dataset has a large number of images, 20% testing data should be sufficient for evaluating the model's performance. This is used in both MLP and SVM models to allow the results to be comparable.

IV. Model Design and Configurations

A. Multi-Layer Perceptron (MLP)

- *Architecture:*

For the architecture of MLP model, I used the keras API to define the neural network. The input layer of the model is a flattened 28 by 28 greyscale images. The first hidden layer is a dense layer with 256 neurons and using the ReLU activation function and with the subsequent layer being half the amount of neuron compared to the first layer. With the output layer having 43 neurons corresponding to the number of categories in the dataset with a softmax activation function. Changing the number of layers, activation functions and the number of neurons in the hidden layers did not significantly improve or worsen the results, though changing the activation function from sigmoid to ReLU did improve the first epoch accuracy but resulted in the same accuracy after 10 epochs. The number of neurons were chosen because it provided a balance between complexity and computation time.

- *Hyperparameters:*

The hyperparameters used for the MLP model is an epoch of 10 and a validation split of 0.2. An epoch of 10 was chosen as it any more epochs after 10 did not provide a meaningful increase in accuracy and a validation split of 0.2 helps monitor the model's performance during training and gives an early indication of overfitting.

- *Optimization:*

The Adam optimizer was used as it effective for a wide range of problems. The Sparse Cross Entropy loss function was used because it is suitable for classification tasks with integer labels.

B. Support Vector Machines (SVM)

- *Kernal Function and Hyperparameters:*

The kernel function for the SVM model is linear with a C of 1. I trained the model with a rbf and linear kernel functions and observed that the accuracy of both are around the same therefore I chose the linear kernel as it is computationally less intensive than rbf due to it being a linear function instead of a non-linear function. The C parameter was kept at 1 as increasing it did not significantly increase the accuracy of the model. Due to a high initial accuracy, I decided not to modify any other parameters, leaving them to their default values.

V. COMPARATIVE ANALYSIS OF MODEL PERFORMANCE

A. Evaluation of Metrics

B. Model Performance

After creating the model and testing it, the metrics seen in Table 1 Model Performance are the results of the testing. The table shows the accuracy, recall, precision and F1 score of the testing.

Model	Accuracy	Recall	Precision	F1-Score
MLP	93.01%	93.01%	93.37%	92.96%
SVM	97.19%	97.19%	97.23%	97.18%

Table 1 Model Performance

As seen in the table, both models received an accuracy of over 90%, the SVM model performs around 4% better than the MLP model at around 97% and the MLP performing at around 93%. Further analysis will be in the points below.

- A high accuracy for both the models suggests that the model correctly classifies most of the images to the correct category. But may not tell the full story if there are class imbalances.
- A high precision value for both the models indicates that the model predicts the correct category for each image therefore minimising on false positives. This means that when the model is predicting a specific category, it is likely to be correct.

- A high recall value for both the models indicates that the model is correctly identifying the correct images that belong in the specific category meaning that it is capturing most of the actual positive instances.
- A high F1-score value for both models indicate that it is detecting true positives and avoiding false positives as it is considering both recall and precision. This value is important as both misidentifying signs and missing signs are both costly especially in real world application such as autonomous driving.

Though there is a high accuracy for both models, within the classification report the values do vary for each category, For the MLP model the precision value does stay above 80% for all the categories but some categories in recall hover in the lower 70% indicating it is not detecting true positives as often as the other categories. The SVM model only has one category with a recall in the lower 70% with the rest of the categories being higher than 80% on all metrics.

VI. CONCLUSION

After evaluating both the MLP and SVM performance, the SVM model performed better across the board compared to the MLP model which indicates that it is better overall. This means that the SVM model is more suitable in real world tasks like autonomous driving systems.

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